

Functional Decomposition and Shapley Interactions for Interpreting Survival Models

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Main Contributions

- ♥ **SurvFD**: A functional decomposition for survival prediction functions that separates effects into **time-dependent** and **time-independent** components.
- ♥ **Theory across survival scales**: Proving that transformations from log-hazard to hazard and survival functions induce **spurious interactions** and **propagate time-dependence**.
- ♥ **SurvSHAP-IQ**: A practical estimator for higher-order, time-indexed Shapley interactions with axiomatic guarantees.
- ♥ **Validation**: Synthetic benchmarks, real-world cancer applications, and a multi-modal histopathology + clinical demonstration.



Paper



Code

Problem: Survival explanation methods **do not capture feature interactions**.

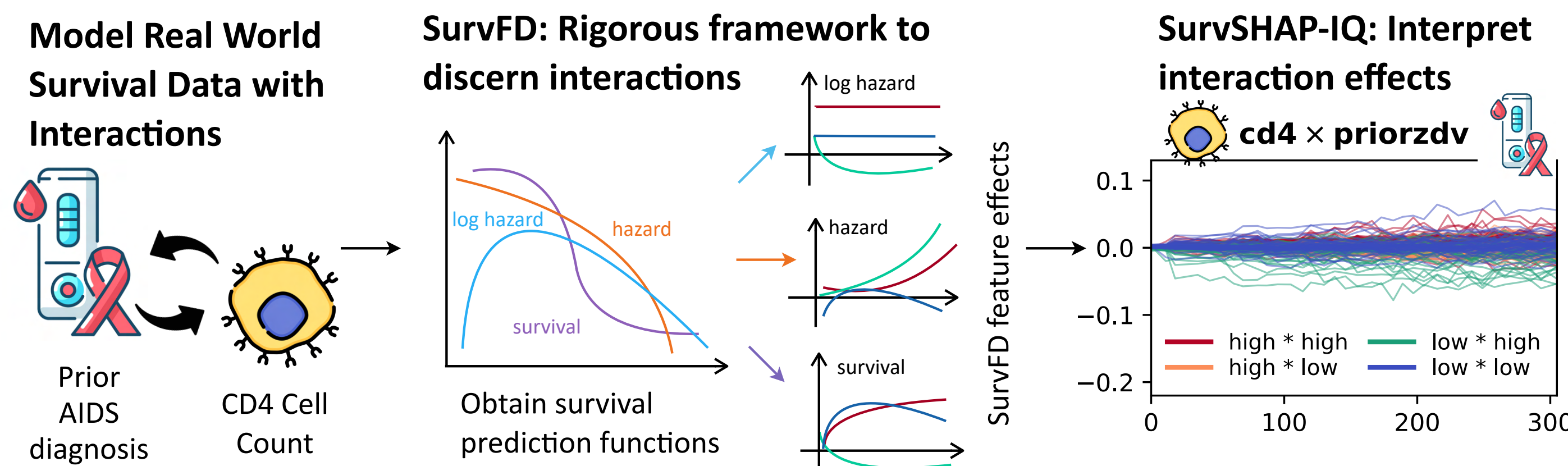


Figure: SurvFD and SurvSHAP-IQ pipeline for interpreting feature interactions in survival models.

Survival Analysis

Survival analysis predicts **time-to-event distributions** for individuals based on their features $x \in \mathcal{X}$. Two key prediction targets are

$$h(t|x) := \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(t \leq T \leq t + \Delta t \mid T \geq t, x)}{\Delta t} \quad (\text{hazard})$$

$$S(t|x) := \mathbb{P}(T \geq t \mid x) = \exp\left(-\int_0^t h(u|x) du\right) \quad (\text{survival})$$

We consider the **multiplicative hazard model**

$$h(t|x) = h_0(t) \exp(G(t|x)),$$

with ground-truth risk score $G(t|x) = \sum_{M \in \mathcal{I}_d} g_M(t|x) + \sum_{M \in \mathcal{I}_{id}} g_M(x)$, generalizing CoxPH to allow **time-dependent**, **non-linear**, and **interaction** effects.

Functional Decomposition

Functional decomposition (FD) writes any square-integrable prediction function $F : \mathcal{X} \rightarrow \mathbb{R}$ as a sum of **pure effects** over feature subsets $M \subseteq P = \{1, \dots, p\}$:

$$F(x) = f_0 + \sum_{\emptyset \neq M \subseteq P} f_M(x), \quad f_M(x) := \int F(x) d\mathbb{P}_{X_{\bar{M}}} - \sum_{Z \subset M} f_Z(x).$$

$|M| = 1$ are **main effects**; $|M| \geq 2$ are **interaction effects**.



How do **feature interactions** interact with **time** in survival predictions?

How can we **estimate** higher-order, time-indexed interactions in practice?

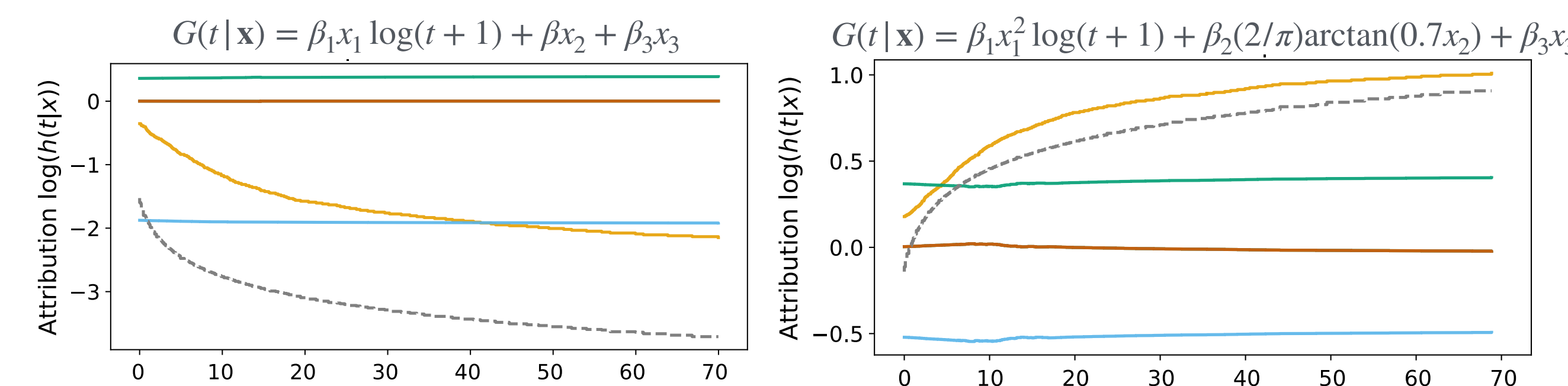
SurvFD: Functional Decomposition for Survival

For any square-integrable survival prediction function $F : \mathcal{X} \times \mathcal{T} \rightarrow \mathbb{R}$:

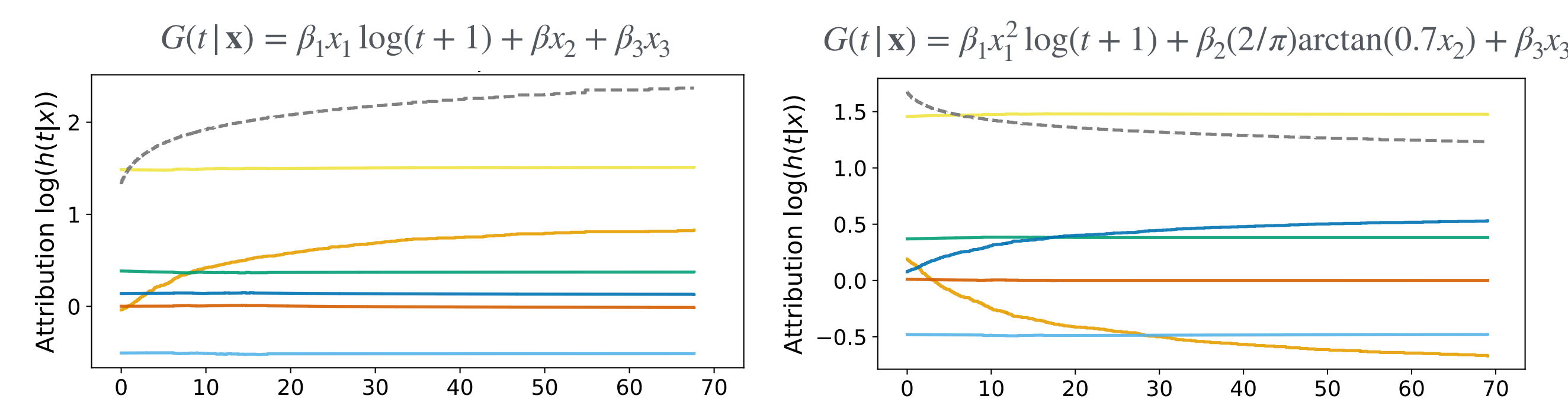
$$F(t|x) = f_0(t) + \sum_{M \in \mathcal{I}_d^*} f_M(t|x) + \sum_{M \in \mathcal{I}_{id}^*} f_M(x)$$

with **time-dependent** ($\mathcal{I}_d^*$) and **time-independent** ($\mathcal{I}_{id}^*$) pure effects.

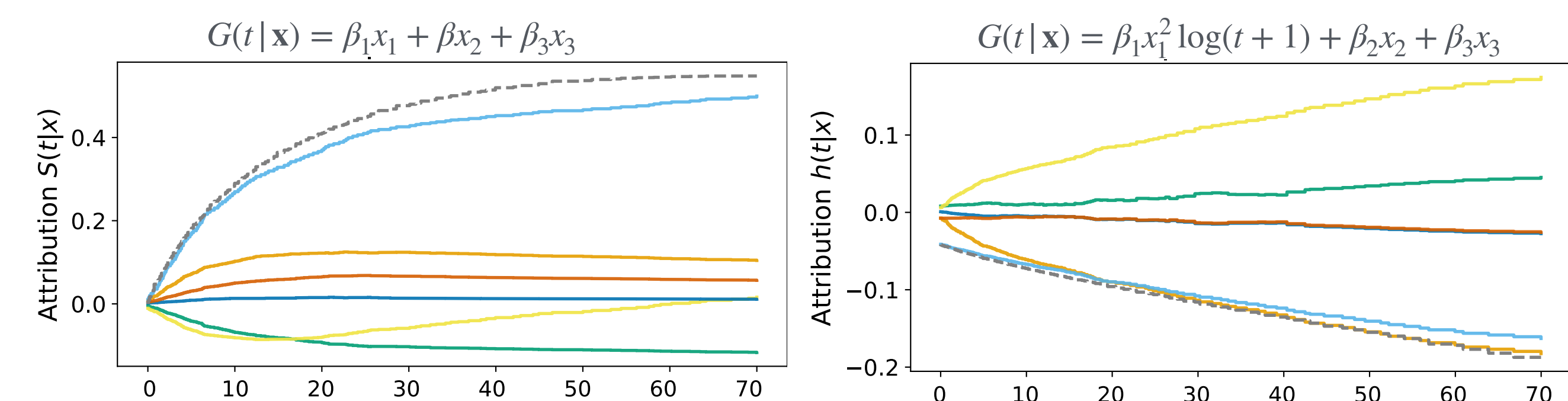
→ **Log-hazard**: SurvFD **recovers** the ground truth for linear or additive $G(t|x)$ (Thm. 3.2).



→ **Log-hazard with TD interactions**: time-dependence **propagates downward** but not upward (Thm. 3.3).



→ **Hazard & survival**: non-linear transforms induce **additional** interactions and **both** downward and upward propagation (Cor. 3.4, Prop. 3.5).



SurvSHAP-IQ: Shapley Interactions for Survival

At any fixed timepoint t , the order- k decomposition is

$$F(t|x) = \sum_{M \subseteq P: |M| \leq k} \phi_M^{(k)}(t|x),$$

where $\phi_M^{(k)}$ with $|M| \geq 2$ are **interaction effects**.

→ **Practical estimator of SurvFD**: $\phi_M^{(k)}(t|x)$ **recovers the pure SurvFD effects** $f_M(t|x)$ exactly at full order; under truncation, higher-order terms are redistributed to lower orders.

→ **Axiomatic**: satisfies symmetry, linearity, dummy, and efficiency at each t .

→ **Marginal or conditional** reference distribution gives **interventional** or **observational** explanations.

→ **Approximators**: regression- & sampling-based (extends SHAP-IQ [1]).

Application: Uveal Melanoma

We explain a **gradient boosting survival model** (validation C-index 0.758) predicting survival in patients with uveal melanoma [2].

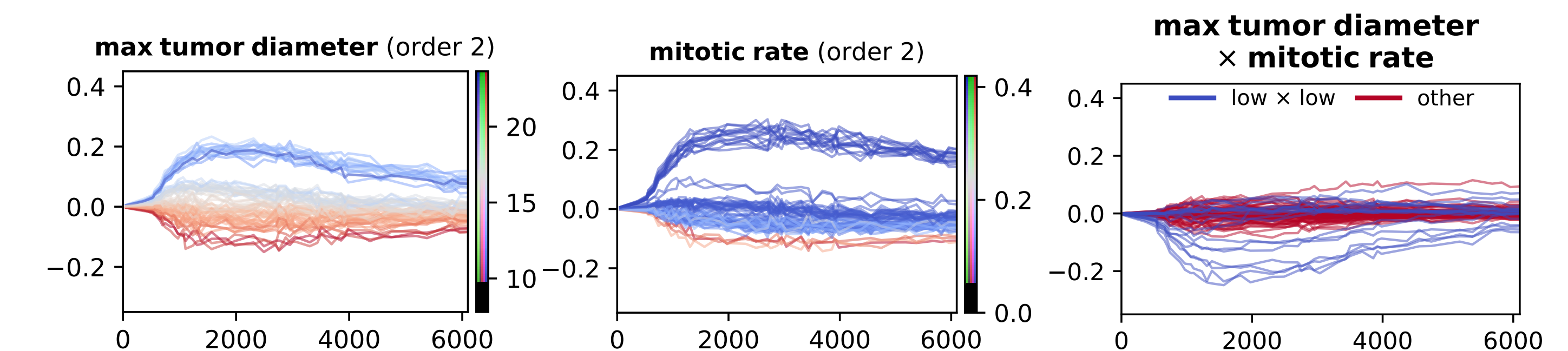


Figure: Order-2 SurvSHAP-IQ effects: maximum tumor diameter and mitotic rate (left, middle) emerge as key predictors, matching established prognostic factors. Their negative pairwise interaction (right) is smaller than the sum of individual effects, indicating partially redundant prognostic information.

→ Second-order effects **disentangle spurious heterogeneity**: variance in the first-order attributions is redistributed into the pure interaction panels.

Application: Multi-modal TCGA-BRCA

We explain a **multi-modal DeepHit model** combining histopathology whole-slide images with clinical features, for breast cancer survival.

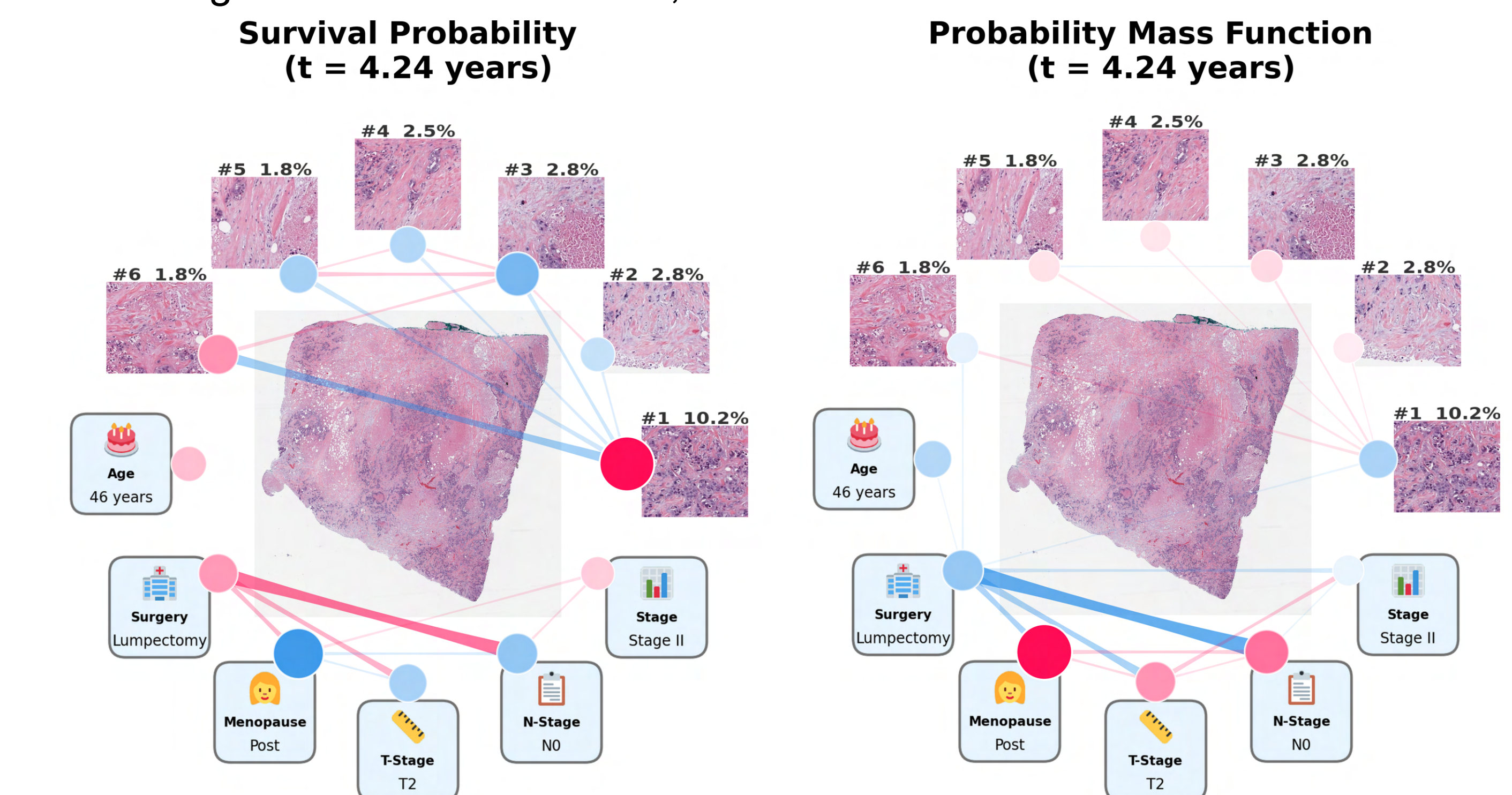


Figure: SurvSHAP-IQ network for a single patient at $t = 4.24$ years. Node size: individual effect. Edges: pairwise interactions. Red: increases survival. Blue: decreases survival.

→ The model recovers the **clinically established prognostic joint effect** of N0 status and lumpectomy [3].

→ Interactions occur **within** rather than **across** modalities.

→ Survival and PMF show **distinct interaction patterns** — empirically extending Prop. 3.5 beyond multiplicative hazards.

References

- [1] Fumagalli, F.; Muschalik, M.; Kolpaczki, P.; Hüllermeier, E.; and Hammer, B. (2023). SHAP-IQ: Unified approximation of any-order Shapley interactions. NeurIPS.
- [2] Donizy, P. et al. (2022). Machine learning models demonstrate that clinicopathologic variables are comparable to gene expression prognostic signature in predicting survival in uveal melanoma. European Journal of Cancer, 174: 251–260.
- [3] Fisher, B. et al. (2002). Twenty-year follow-up of a randomized trial comparing total mastectomy, lumpectomy, and lumpectomy plus irradiation for the treatment of invasive breast cancer. NEJM, 347(16): 1233–1241.

